

Examining Road Freight Transport Costs: A Philippine Perspective

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Abstract: The cost of road freight transport in the Philippines is perceived to be higher than its neighboring countries. Additionally, there is limited literature exploring the magnitude of this issue.

This study serves as a practical effort to quantify and analyze the high transport costs in the Philippines using comprehensive empirical data. It utilizes information from a Luzon-wide survey of 36 trucking and manufacturing companies operating along the country's main logistics corridors. Key findings reveal that at 9.7 US cents per ton-km, the cost of the transport of goods in the Philippines is among the highest within Southeast Asia. Further analysis indicates a possible correlation between the cost of road freight and the quality of road infrastructure, particularly the level of development of the expressway network. The results of this study aim to guide planners and policymakers in the Philippines to take the necessary steps to increase the country's trade competitiveness.

Keywords: road freight, logistics cost, expressways, benchmarking, logistics performance index, global competitiveness index

1. INTRODUCTION

The Philippine economy relies on the logistic transport of goods throughout the country. Although it is an archipelagic country, it has been observed that a majority of logistic operations are still dependent on roads for inland freight transport.

At present, manufacturing companies in the country are faced with the challenge of high logistic costs which are perceived to be even higher than similar neighboring countries. Due to the limited literature exploring the magnitude of this problem, it is imperative that the country's logistics performance be objectively placed into perspective by comparing it with the logistics cost of other countries with similar situations.

In order to better understand why and by how much the logistics cost in the Philippines is higher than its neighboring countries, surveys were conducted with major manufacturing firms located in Special Economic Zones (SEZ) all over the country. Additionally, trucking firms were also surveyed to further develop quantitative and accurate findings as to the flow of cargo to and from the SEZs to any major destination in Luzon such as ports, airports, and distribution centers.

In essence, this study aims to quantify and analyze the cause for the high transport costs

of road freight in the Philippines. It also serves to determine the correlation between road freight costs and road infrastructure with a focus on the development of the expressway network. If a statistically significant correlation is established, it may be deduced that specific measures such as the development of road infrastructure will improve the country's logistics performance and trade competitiveness as a whole.

2. BACKGROUND

2.1 Relationship between logistics cost and road infrastructure

The impact of related infrastructure on logistics costs is undeniable. The costs incurred to deliver goods and services from production zones (e.g., manufacturing companies) to seaports/airports are heavily dictated by the availability and quality of transport infrastructure. This is considered to be significant especially to developing countries. According to the World Development Report, more than 50% of firms in Latin America and Africa consider infrastructure a serious hurdle in logistics (World Bank, 2005).

Several studies have previously explored the effects of infrastructure on logistics costs. Calderon and Servén (2002) used empirical data to discuss how Latin America's poor road infrastructure raises transport and logistics costs. Using navigational data in five German regions, Kellner, *et al.* (2016) found out that freight costs tend to be higher in areas with less favorable transport infrastructure. Tolliver and Dybing (2007) explored the effects of improved road quality (higher load limit) on lowering the unit transport costs of farmer-owned grain trucks. Guasch (2001) stated that poor infrastructure, such as roads and ports, is one of the primary determinants in high inventory and logistics costs in developing countries.

2.2 The trucking system in the Philippines

The trucking industry plays a large role in the Philippines' logistics system. Several issues continue to surround the industry especially in more dense areas like Metro Manila which they need to pass through to get to the warehouses or distribution centers. Table 1 summarizes the governing laws with regard to the trucking system in the Philippines.

Table 1. Governing Laws of the Trucking System

Governing Law	Description
Republic Act 4136	Land Transportation and Traffic Code
Republic Act 8794	Penalty for Overloading
MMDA Truck Ban Ordinance	Truck Ban
DOTr Department Order No. 2017-009	15-year Age Restriction on Trucks
LTFRB Memorandum Circular No. 2018-007	Roadworthiness Tests
Commonwealth Act No. 146	Public Service Law

Note: MMDA = Metro Manila Development Authority;

DOTr = Department of Transportation;

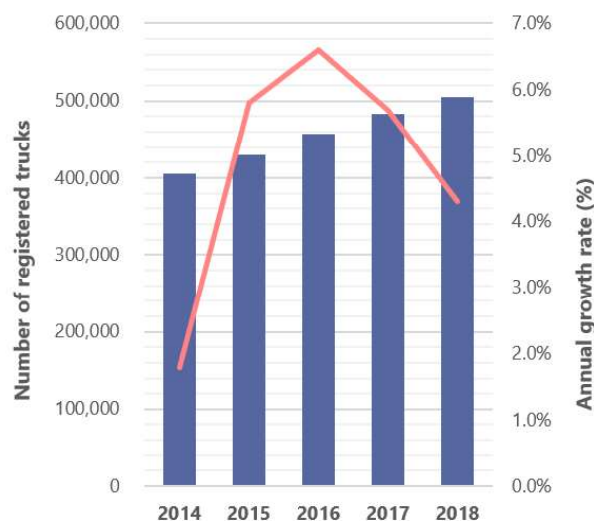
LTFRB = Land Transportation Franchising and Regulatory Board

Based on the Republic of Philippines, Republic Act No. 4136 "Land Transportation and Traffic Code", there is a maximum allowable Gross Vehicle Weight (GVW) of 41,500 kg for

truck semi-trailers with four axles (14 wheels) and 42,000 kg for truck-trailers with three axles at motor-vehicle and three axles at trailer (22 wheels). It is worth noting that the study this computation was based on was only able to derive the maximum single axle load and was not able to estimate maximum loads for tandem and tridem axles. Tandem axles are two axles positioned one after the other while tridems are three axles grouped together. These tandems and tridems are typical configurations for the rear axles of large trucks and trailers, enabling them to support heavy loads that are typically more distributed towards the rear axles.

There also exists a corresponding penalty for overloading which is known as Republic Act No. 8794, stating that the Motor Vehicle User's Charge (MVUC, in lieu of RA 4136 s.8) shall be imposed on trucks and trailers for loading beyond their prescribed gross vehicle weight provided that no single axle load shall exceed 13,500 kg.

According to a study by Bacero and Fillone (2003), "there is a significant increase of truck volume observed in the circumferential and radial road of Metro Manila and congestion in the truck routes were evident based on the long queue of trucks observed on the major thoroughfares." The data from the Land Transportation Office indicated that the number of registered trucks and trailers grew at an average annual rate of 5.6% from 2014 to 2018 (Figure 1).



Source: Land Transportation Office (2019)

Figure 1. Annual growth rate of trucks and trailers in the Philippines

Bacero and Fillone (2003) have also stated that "based on the review of truck policies in Metro Manila, truck ban time were varied and uncoordinated across cities." The truck ban, which is another governing ordinance in the Philippines, prohibits trucks from using main thoroughfares at specific times of the day to avoid traffic congestion. A study by Patalinghug, *et al.* (2015) revealed that three out of seven locators experienced doubled truck rates following the implementation of the truck ban. Furthermore, Llanto (2016) estimated an economic loss of PhP 43.85 billion over a seven-month cargo truck ban in 2014.

The worsening traffic congestion in Metro Manila and other major cities where trucks have to pass to reach their destination contributes significantly to the high logistics costs (Rodolfo and Antonio, 2006). With an annual increase in the volume of trucks and having to conform to the strict policies, trucking companies continue to struggle in the timely and effective delivery of goods.

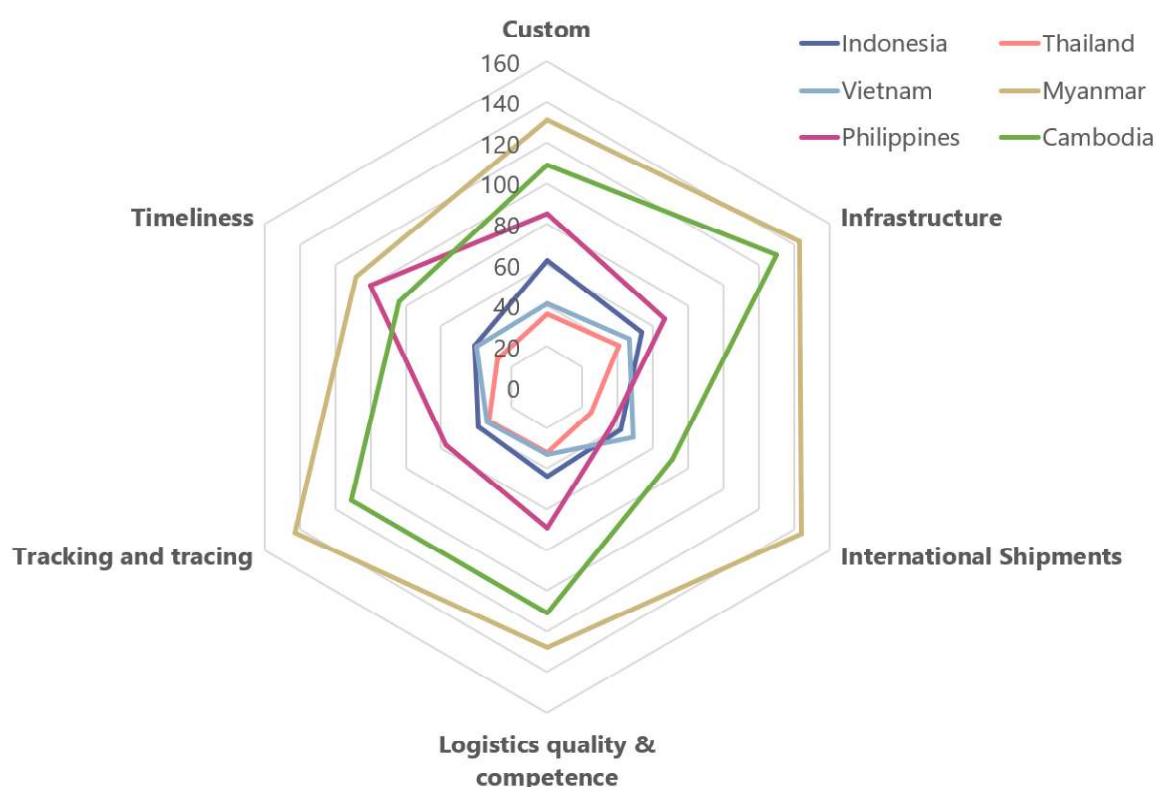
2.3 LPI of the Philippines vs. other ASEAN countries

To identify the challenges and opportunities countries are facing in terms of their logistics and trade performance, the World Bank (WB) created a benchmarking tool, which compares the Logistics Performance Index (LPI) of 160 countries around the world.

The Logistics Performance Index (LPI) is the weighted average of the country scores with regard to these six (6) key dimensions:

- 1) Efficiency of the clearance process (i.e., speed, simplicity and predictability of formalities) by border control agencies, including customs;
- 2) Quality of trade and transport related infrastructure (e.g., ports, railroads, roads, information technology);
- 3) Ease of arranging competitively priced shipments;
- 4) Competence and quality of logistics services (e.g., transport operators, customs brokers);
- 5) Ability to track and trace consignments;
- 6) Timeliness of shipments in reaching destination within the scheduled or expected delivery time.

The LPI of the Philippines in comparison with other ASEAN countries is put into detail in Figure 2. A lower value corresponds to a more favorable logistics performance indicator.



Source: Logistics Performance Index (LPI) Report, World Bank (2018)

Figure 2. Logistics performance of the Philippines compared to other ASEAN countries

It can be observed from the figure that the Philippines scored 5th out of 6 in terms of timeliness and placed 4th out of 6 in terms of its logistics infrastructure. This explains that there exists the need for an improvement in transport related infrastructure such as ports, railroads, roads, and information technology.

With road infrastructure being the dominant mode of inland transportation, an improvement in such will also increase reliability thus affecting timeliness. An enhancement of both factors may definitely increase the country's overall LPI.

2.4 Logistics Cost comparison among ASEAN countries

The Department of Trade and Industry (DTI) and World Bank (WB) jointly undertook a study in 2017 with the objective of defining the logistics cost in the Philippines as compared to the logistics cost of other ASEAN countries. The study revealed that the Philippines has the highest logistics cost among the four (4) ASEAN countries as illustrated in Figure 3.

The Philippines has the highest share in all the logistics cost's sub-components: transport, warehousing, inventory, and logistics administration. With the share of the transport cost in the country almost doubling in comparison to Thailand, it is imperative that road infrastructure be focused on in order to improve.

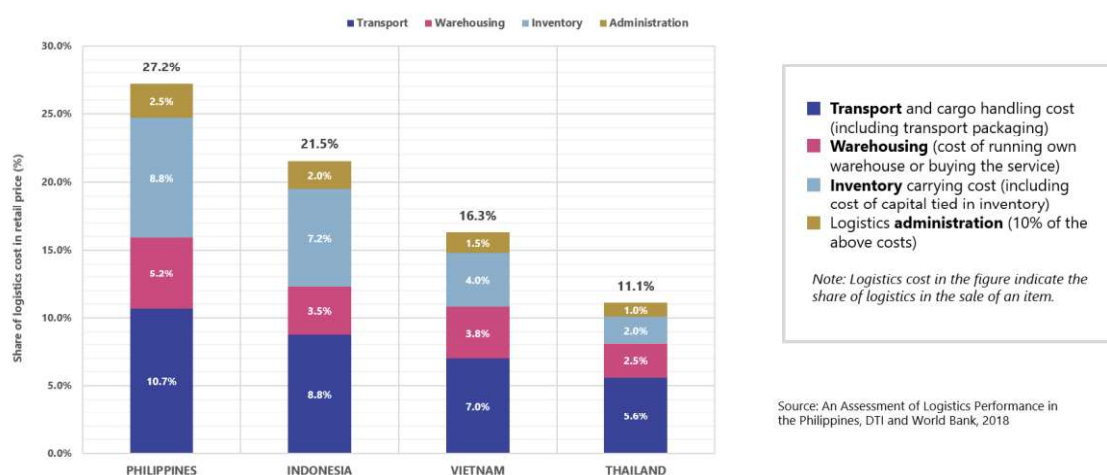


Figure 3. Logistics cost in the Philippines compared to other ASEAN countries

In this study, DTI and World Bank (2017) identified that from users' perspective, the most important issues are as follows: reliability (44%), cost (30%), and time (26%). Reliability, as the most significant, may again be improved by providing an effective and efficient road network that links different logistics facilities (e.g., ports, airports, factories).

Furthermore, the study also implied that due to the low reliability of the logistics system (e.g., unpredictable arrivals due to customs related problems, traffic congestions, etc.), manufacturers are forced to maintain a large volume of supply which require securing huge warehouses and thus calls for good inventory and administration. This in turn requires additional cost and could partly explain the reason for the high logistics costs in the Philippines.

If an effective road network is put into place, this will secure the timely delivery of goods which in turn also affect the other sub-components of the logistics cost such as warehousing, inventory, and logistics administration. This is anchored on the fact that having a more reliable logistics system will prevent overstocking hence reducing their need to maintain a huge inventory and the need for bigger warehouses. With a reliable logistics system, manufacturing companies will also have an easier time planning efficiently, bringing down logistics cost.

2.5 Logistics cost comparison among major islands of the Philippines

Disaggregating the logistics cost data among the major islands of the country, it can be observed that there is a huge discrepancy among them. Figure 4 demonstrates this difference, with the major island of Mindanao having the highest logistics cost and Luzon performing the best among the three.

According to DTI and World Bank (2017), “Mindanao appears to face the highest logistics cost/sales, reaching almost a third of its sales, while Luzon face the lowest. It is also worthwhile to note that Luzon's logistics cost/sales are lower than that of the national average. Luzon is the main industrialized island group in the country and thus, benefits from lower logistics cost, consistently in all components. Cost components in Mindanao, meanwhile, emerge as the highest among the three island groups.”

As for the high transport cost in Visayas, the study indicates that this is most likely attributed to the geographical nature of the region, with a multitude of inter-island shipping used in the logistics of freight.

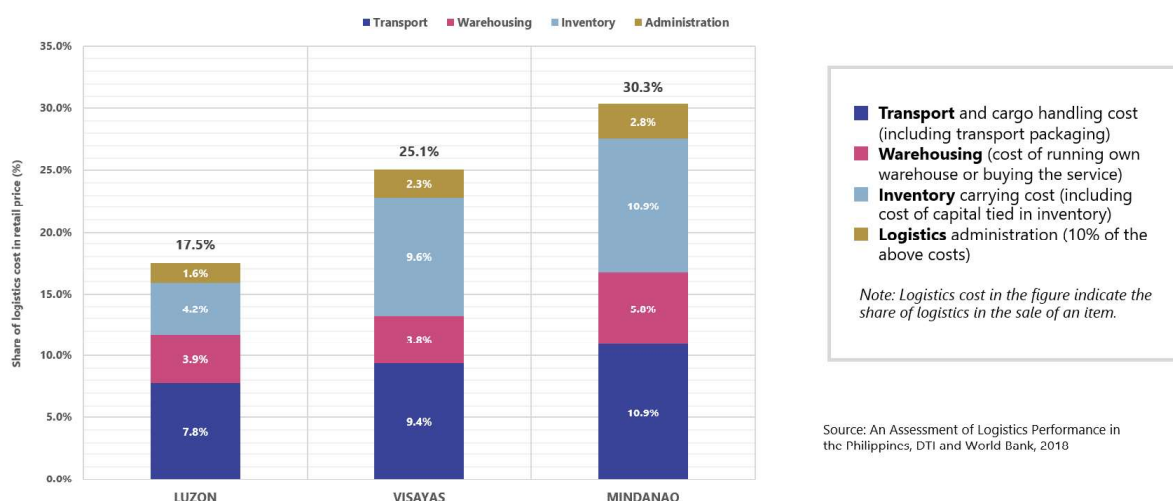


Figure 4. Logistics cost comparison among major islands of the Philippines

The preceding studies have established the fact that the Philippines has high logistics costs, but it was not directly quantified. As such, the study was conducted to compliment this lack of quantified data through the use of surveys.

3. SCOPE AND LIMITATIONS

Due to the efficiency of logistics operations and availability of data in Luzon, this study only surveyed logistics and manufacturing companies located in or operating in Luzon. Despite the expected higher logistics cost in Visayas and Mindanao, there is more comprehensive data, which is representative of the Philippines, in Luzon. This also holds true since multi-national companies with more complete and accurate trucking data are located here.

4. METHODOLOGY

4.1 Overview

Two types of surveys were conducted for this study which were targeted at two sectors: the logistics companies and the manufacturing companies. Table 2 describes the objectives of each survey.

Table 2. Types of surveys conducted

Type of Survey	Objectives
Logistic company interview survey	To get information on the type of commodities handled by the trucking company and understand how these products are transported
Manufacturing company interview survey	To understand the flow of raw materials and manufactured goods within the target company; to determine the routes undertaken by the transport vehicles

4.2 Survey Area

A total of six (6) logistics companies were selected in Luzon with specific locations at Metro Manila, Region III, and Region IV-A.

Whereas for the manufacturing companies, a total of thirty (30) were selected from ten (10) Special Economic Zones (SEZ) as listed in Table 3. The location of each Special Economic Zone can be found in Figure 5.

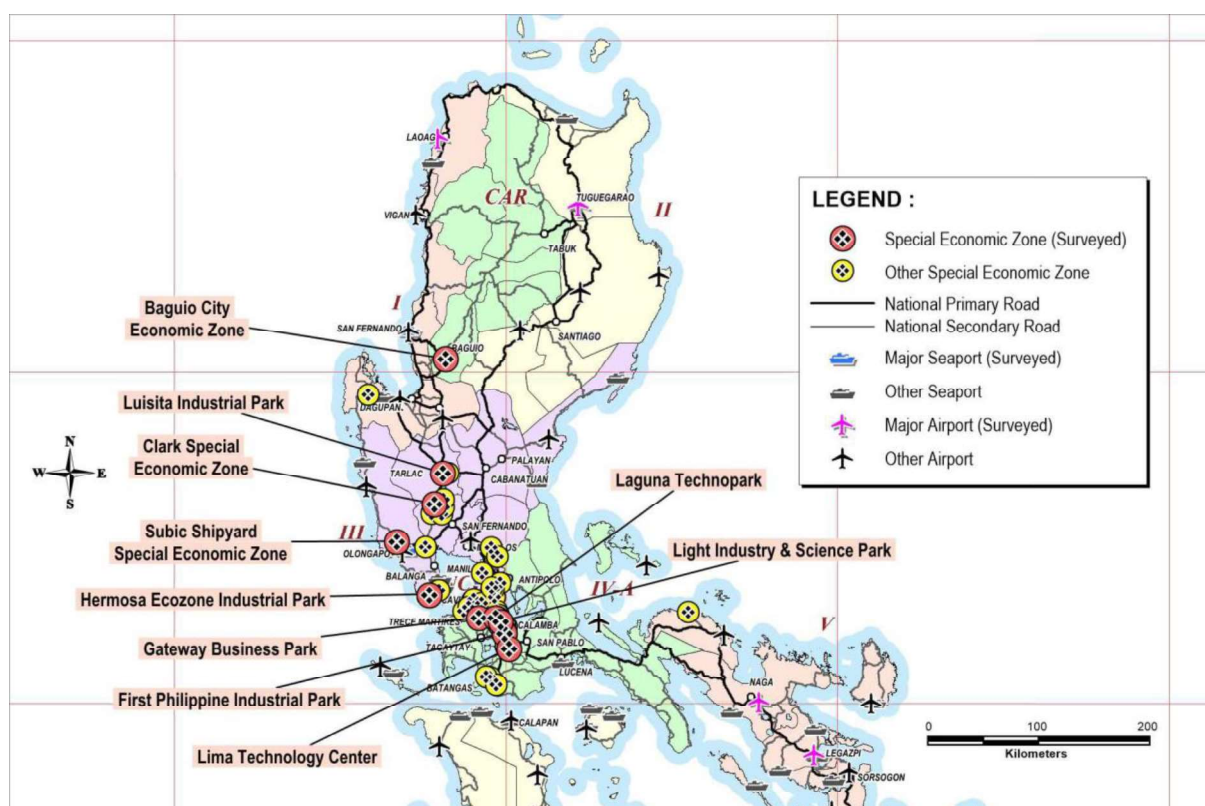


Figure 5. Map of Special Economic Zones

Table 3. Location of manufacturing firms covered by the survey

No.	SEZ	Location	Number of Firms Interviewed
1	Baguio City Economic Zone	Baguio City, Benguet	4
2	Luisita Industrial Park	San Miguel, Tarlac	1
3	Clark Special Economic Zone	Angeles, Pampanga	2
4	Subic Shipyard Special Economic Zone	Subic, Zambales	4
5	Hermosa Ecozone Industrial Park	Hermosa, Bataan	2
6	Gateway Business Park	General Trias, Cavite	4
7	First Philippine Industrial Park	Sto. Tomas, Batangas	1
8	Laguna Technopark SEZ	Binan and Sta. Rosa City, Laguna	6
9	Light Industry & Science Park	Cabuyao, Laguna	1
10	Lima Technology Center	Lipa, Batangas	5
Total			30

4.3 Survey Items

All surveys were conducted through a face-to-face interview using a prepared questionnaire of 5 to 6 pages. For both surveys, the surveyor asked the logistics administrator for information on the logistics facilities, volume of commodities, and transport operations of the company. The survey also covered the major problems encountered by the company in transporting cargos and the corresponding recommendations. The survey items for each type of survey can be found in Table 4 and Table 5.

Table 4. Logistics company interview survey methodology

Survey day	One regular weekday (from Monday to Friday)
Survey time	Company office hours
Survey items	Cargo volume per commodity, loading condition, vehicle inventory, warehouse inventory, service area, origin and destination of trucks, staying time of commodity

Table 5. Manufacturing company interview survey methodology

Survey day	One regular weekday (from Monday to Friday)
Survey time	Company office hours
Survey items	Warehouse inventory, production rate per commodity, products exported or consumed, origin of raw material, destination of products, delivery cost of products

5. SURVEY RESULTS

5.1 Logistics corridor in Luzon

The logistics corridor in Luzon, as shown in Figure 6, was identified using the results of the trucking and manufacturing company surveys. The study found out that the location of the SEZs and the routes undertaken by trucks are mostly dictated by the expressway network. Trucks serving the SEZs in Northern Luzon operated along the North Luzon Expressway (NLEX), Tarlac-Pangasinan-La Union Expressway (TPLEX) and the Subic-Clark-Tarlac Expressway (SCTEX). In addition, a segment of the Harbor Link of NLEX opened in 2019 and is expected to partially redirect trucks going to/from the Port of Manila.

In the south of Manila, most trucks preferred to use the South Luzon Expressway (SLEX), Southern Tagalog Arterial Road (STAR) Tollway, Manila-Cavite Expressway (CAVITEx) and the NAIA Expressway (NAIAx) over local roads. The Governor's Drive, which serves as a strategic link between the SEZs in Cavite and the SLEX, was also observed to handle large volume of trucks. The following were observed from the results of the survey:

- Majority of the SEZs involved in the production of light industries (e.g., Baguio City Economic Zone, Clark Freeport Zone, and Gateway Business Park) used airports as the primary gateway for the export of products. Light industries include electronic parts, computer chips, light garments, and products made of plastics.
- SEZs hosting heavy industries (e.g., Luisita Industrial Park, Lima Technology Center, and First Philippine Industrial Park) primarily used ports for the shipment of products. Heavy industries include automobile parts, precision machines, chemicals, and processed food.
- Majority of the import and export activities in Luzon were still heavily concentrated in Manila, mainly due to its efficiency and wide coverage. SEZs such as the Baguio City Economic Zone and the Clark Freeport Zone are situated closer to the Clark International Airport, but still opted to ship their products through the NAIA. Similarly, several companies in the Luisita Industrial Park and the Lima Technology Center utilizes the Port of Manila, instead of the nearby Port of Subic Bay and Port of Batangas.

5.2 Travel time and cost

The trucks' travel time and cost have been studied by comparing travel data in 2015 and 2019. In general, most routes experienced increased travel time and cost, except for those where expressway development took place. The following were the major observations:

- **Impact of TPLEX to SEZs in the North** – Trucks serving SEZs in the far north, particularly the Baguio City Economic Zone experienced improved travel time to/from the gateway ports and airports (e.g. NAIA, Clark International Airport and MICP). This can be attributed to the extension of the Tarlac-Pangasinan-La Union Expressway (TPLEX). The 5.4-km. Urdaneta-Binalonan segment opened to the public in 2016, and the 10.10-km. Binalonan-Pozorrubio segment began operations in 2017.
- **Impact of NAIAx to SEZs in Cavite** – SEZs located in Cavite and its surrounding areas experienced a 30-minute time saving in shipping out their products to/from NAIA (40-km. distance). The shorter travel time was primarily due to the opening of the ramp which connects the NAIAx and CAVITEx. The ramp facilitated the transfer of trucks from one expressway to another without passing through local roads.
- **Impact of SLEX to SEZs in Laguna** – There is a high concentration of SEZs in Laguna and most have been observed to rely on the SLEX for the transport of goods. The increase in congestion along SLEX and in the local roads connecting the SEZs and the expressway had slowed down the travel of trucks. In general, travel time from these SEZs to Manila (and vice versa) increased by at least one hour from 2015 to 2019.

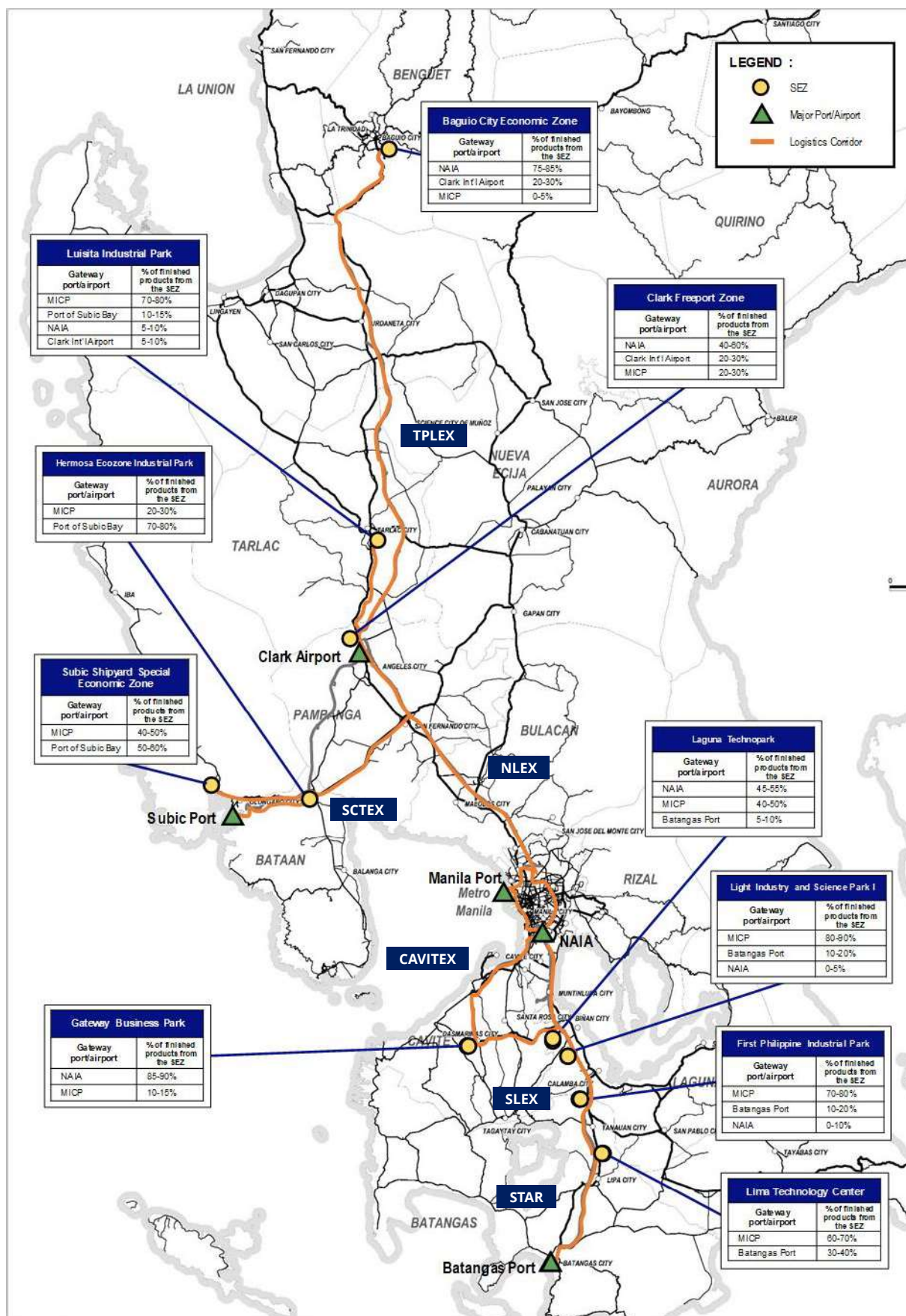


Figure 6. Logistics corridor and SEZ gateway ports and airports in Luzon

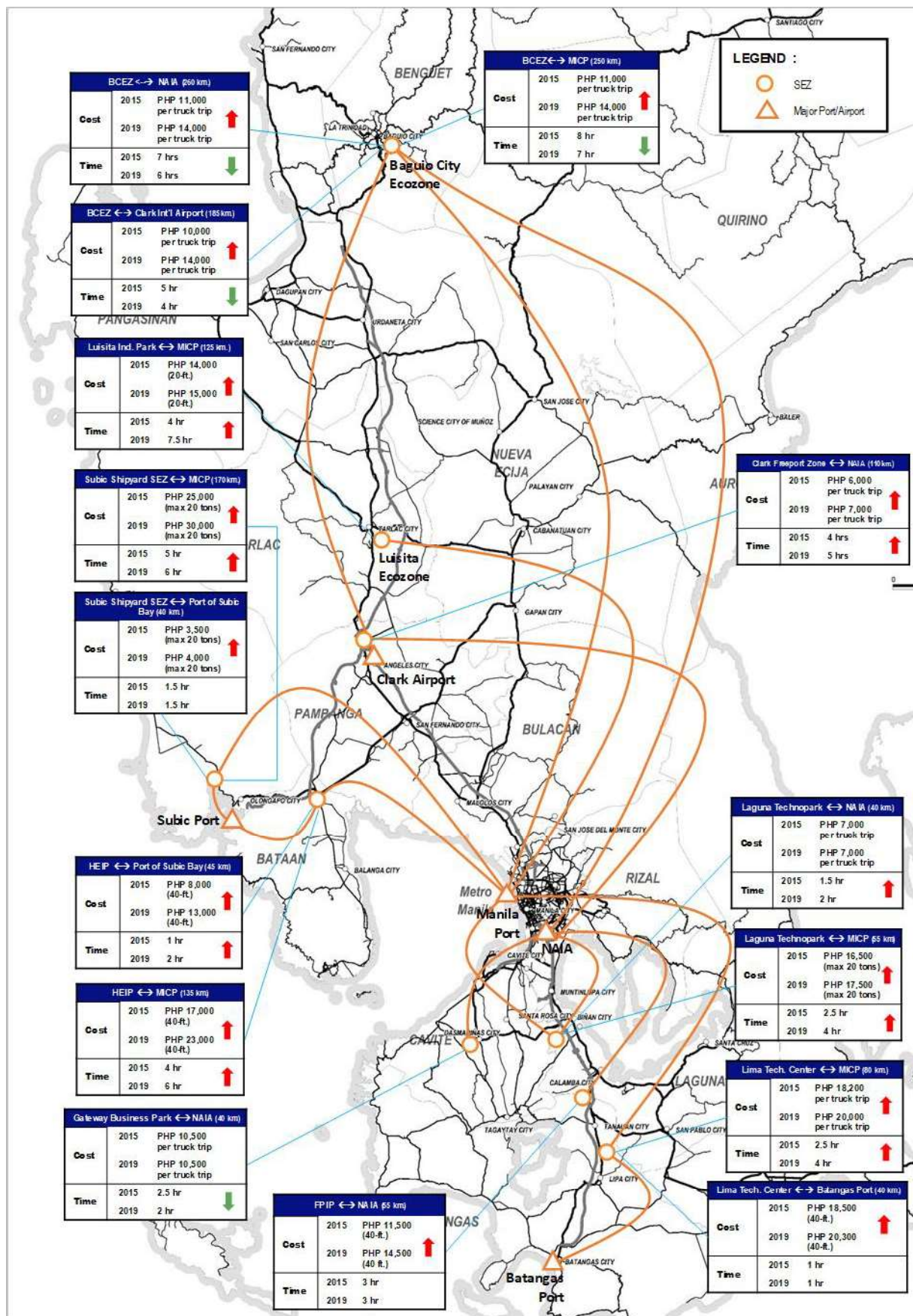


Figure 7. Freight transport cost and travel time in 2015 and 2019

6. ANALYSIS OF ROAD FREIGHT TRANSPORT COSTS

6.1 Unit road transport cost in Luzon

The unit road transport cost was calculated using survey data (distance, commodity weight, cost) provided by 36 trucking and manufacturing firms. The survey covered 45 different truck routes in Luzon hosting a wide array of commodities. The computed value for the unit trucking cost among all surveyed routes in 2019 was 9.67 US cent per ton-km. The cost of road transport increased by 1.4% per annum from 2015 to 2019.

Table 6 shows that the cost of trucking in Southern Luzon is higher compared to Northern Luzon. This could be attributed to the difference in the development of expressways, which are utilized in 96% of all surveyed routes. The expressway system in the north covered a greater distance and recorded less traffic volume compared to the south. Congestion has been observed along the interchanges of the SLEX and several manufacturing firms (4% of surveyed routes) opted not to use the expressways in the South due to the negligible difference in travel time.

The highest transport cost was recorded in the Lima Technology Center $\leftrightarrow$ Batangas Port route (9.8-km.). In this route, trucks had to pass through the STAR Tollway, which converges into a two-lane expressway (one lane per direction) near its southern terminus, thus making it highly vulnerable to traffic congestion.

Table 6. Summary of the unit road transport costs in Luzon

	Road transport cost (US cent per ton-km)	
	2015	2019
Average of all routes	9.19	9.67
Minimum (455 – km. stretch in Northern Luzon)	6.28	7.00
Maximum (9.8 – km. stretch in Southern Luzon)	31.82	40.06
Average of routes in Northern Luzon	5.63	7.15
Average of routes in Southern Luzon	18.71	20.44

Figure 8 shows that the computed unit transport cost varies depending on the length of the truck route. Shorter travel distances are often associated with lower truck load, which results in a higher unit transport cost. In addition, lower average travel speeds were computed in shorter routes. This can be attributed to the traffic congestion along local roads, particularly near Metro Manila, which in turn increases labor and fuel costs.

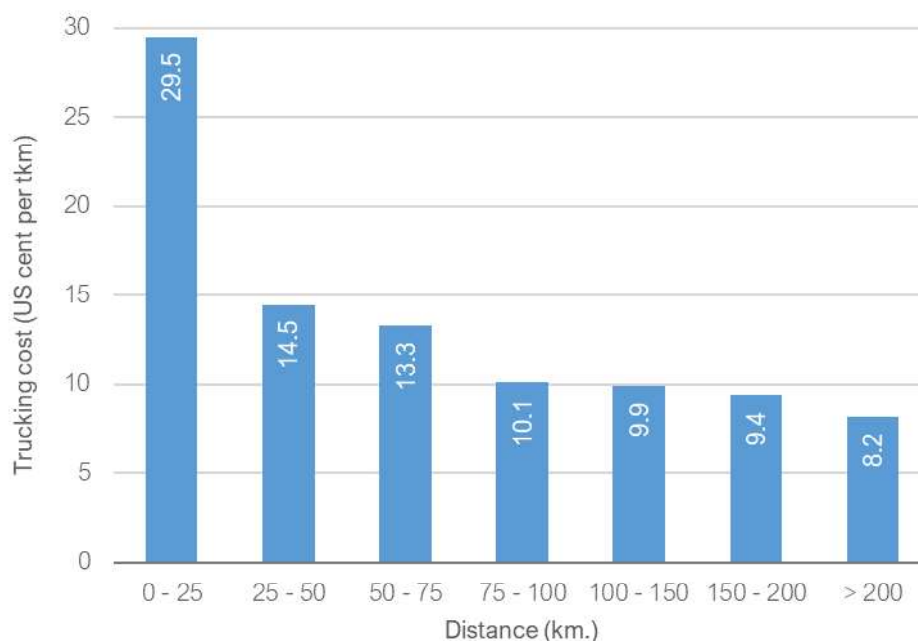


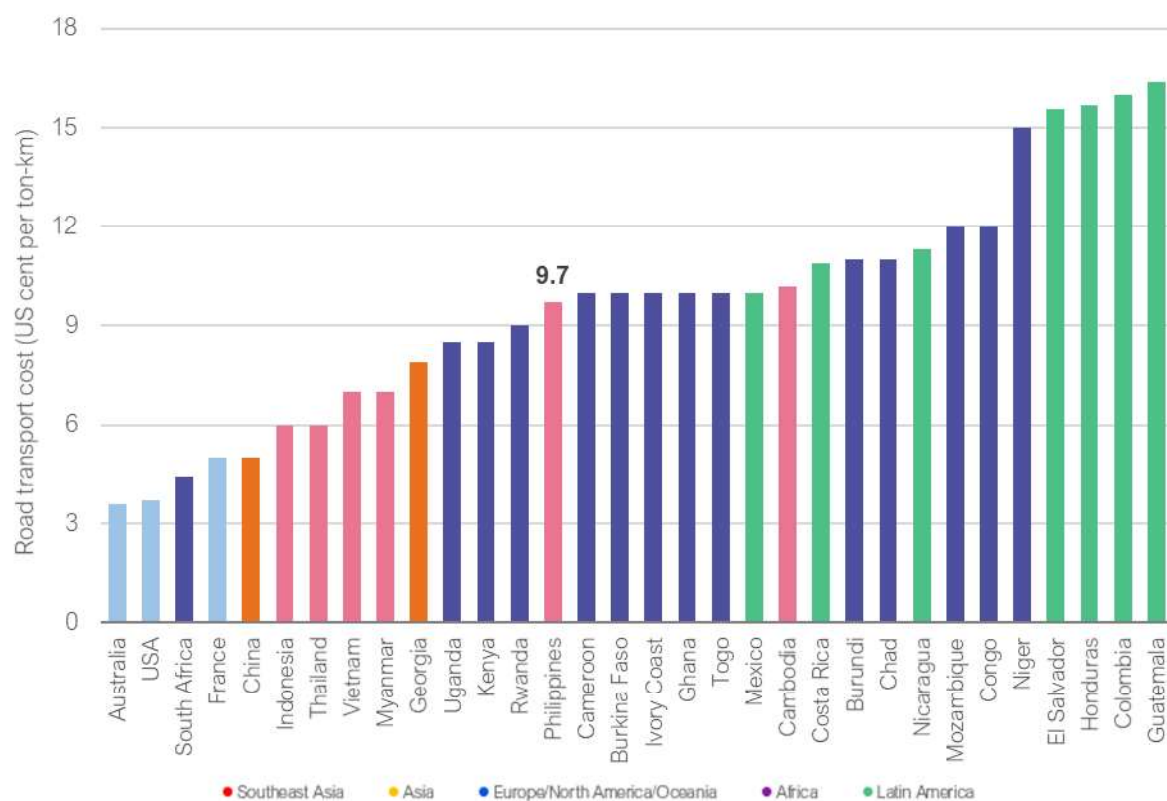
Figure 8. Unit road freight cost per distance of truck route

6.2 International comparison of road freight prices

This study used data from several reports published by the World Bank (WB), Asian Development Bank (ADB), Japan International Cooperation Agency (JICA) and the United States Agency for International Development (USAID) to compare road freight prices among different countries. Figure 9 shows the international position of the Philippines, while Figure 10 presents the level of dispersion of transport costs among different regions.

According to the Asian Development Bank (2012), intercity transport costs within middle-income countries predominantly ranged from 4 to 6 US cent per ton-km. Therefore, the road freight costs in the Philippines can be considered high. At 9.67 US cent per ton-km, it is comparable with the high transport costs associated with landlocked countries in Africa such as Burkina Faso (10 US cent per ton-km), Burundi (11 US cent per ton-km), and Congo (12 US cent per ton-km).

Figure 11 reduces the comparison of road freight transport costs to several ASEAN countries. This indicates that the Philippines is among the least performing countries within the region, along with Cambodia (10.2 US cent per ton-km). The findings of this study complement the results of DTI and World Bank (2018), which stated that the Philippines had a higher share of logistics costs in the retail value of products, as compared to Indonesia, Vietnam and Thailand.



Note: The trucking cost in the Philippines was calculated using the results of the survey by the JICA Study Team (2019). Figures representing the unit transport cost of other countries were taken from studies conducted by the World Bank, Asian Development Bank, Japan International Cooperation Agency, and USAID.

Figure 9. International comparison of road freight costs

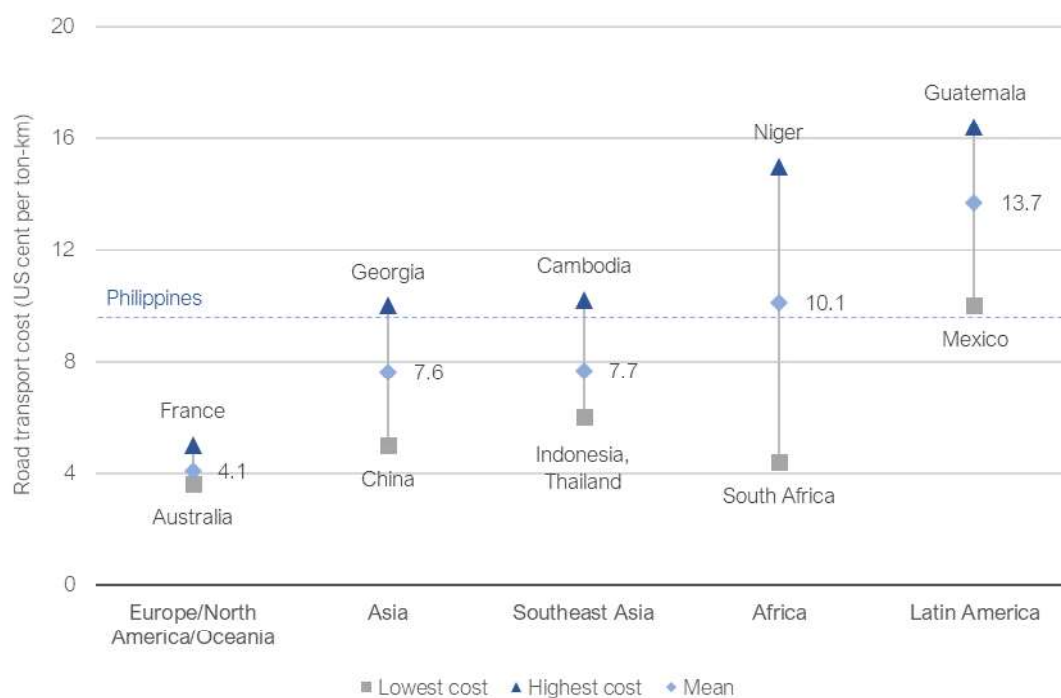


Figure 10. Transport costs depending on regional classification

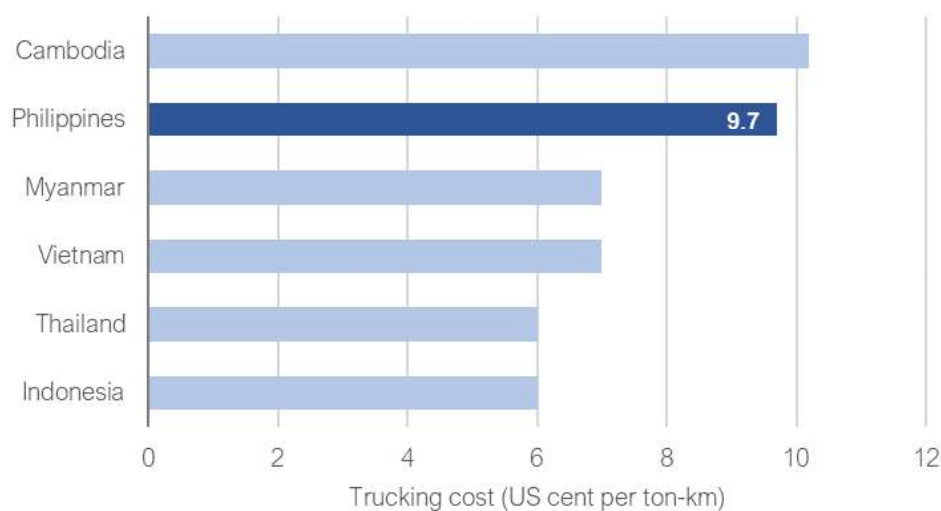


Figure 11. Road freight costs among ASEAN countries

6.3 Relationship between road transport costs and infrastructure

6.3.1 Case 1: Road freight costs vs LPI (World Bank)

The road freight costs of each country were plotted against logistics infrastructure indicators to determine if there is a direct link between the two factors. The first indicator is the Logistics Performance Index (LPI), which is an international benchmarking tool annually published by the World Bank. The LPI makes use of six main indicators, including infrastructure.

As shown in Figure 12, there is an observed clustering among countries with similar geography and/or size of economy. High-income countries such as the United States, France and Australia are all characterized by adequate logistics infrastructure and low transport costs. In contrast, Latin American countries, which are impacted by heavy traffic congestion and high informal costs (World Bank, 2014), were identified as the worst performing cluster.

The Philippines, along with Cambodia, has a level of infrastructure and road freight costs at par with landlocked African countries. Its position in the graph indicates that the Philippines is relatively underperforming compared to other ASEAN countries such as Thailand, Vietnam, and Indonesia. These three countries were likewise identified among the ten overperforming non-high-income countries in the world (World Bank, 2018).

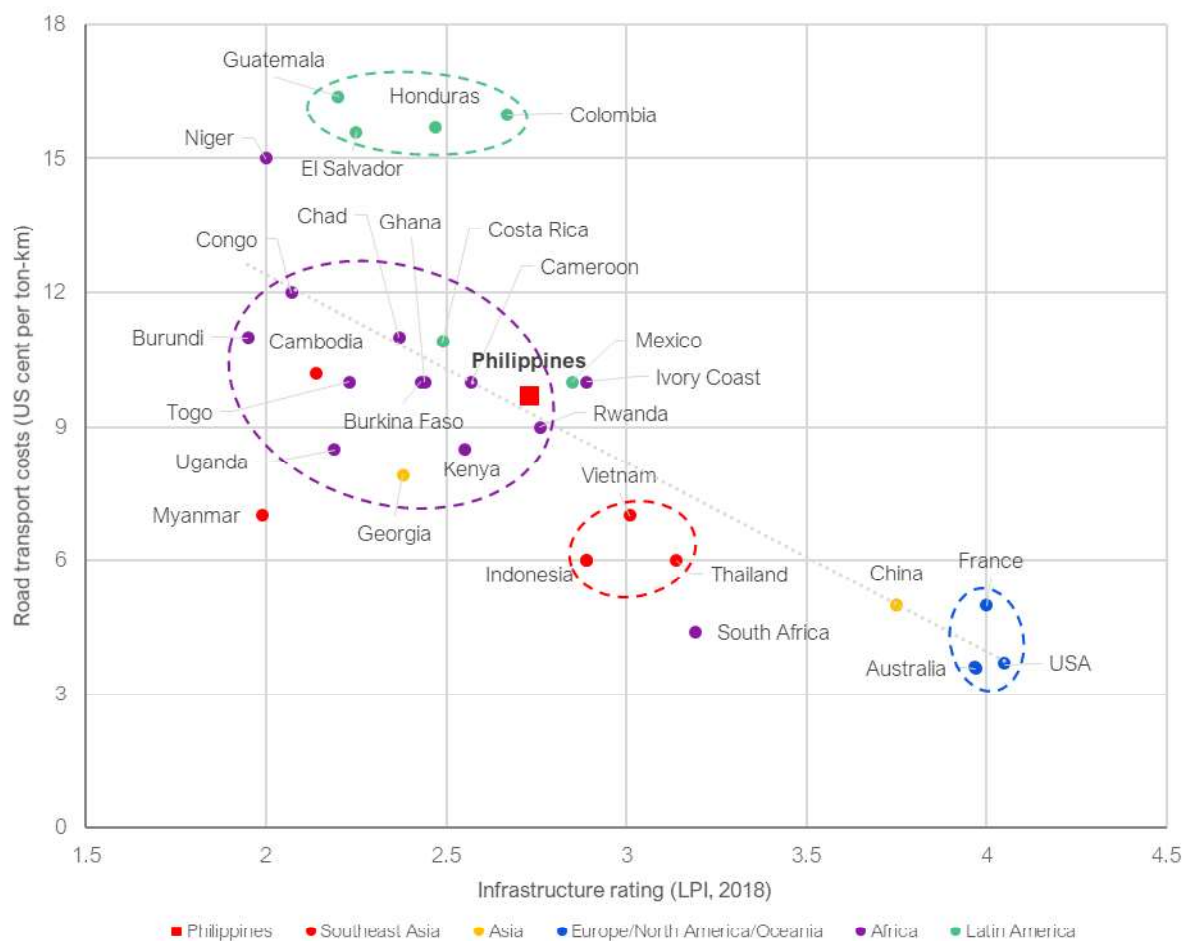


Figure 12. Relationship between road transport costs and infrastructure rating (LPI)

Table 7. Correlation between road transport costs and infrastructure rating (LPI)

		Transport cost	Infrastructure rating (LPI)
Transport cost	<i>Pearson Correlation</i>	1.00	-0.695**
	<i>Sig. (2-tailed)</i>		0.000
	<i>N</i>	30	30
Infrastructure rating (LPI)	<i>Pearson Correlation</i>	-0.695**	1.00
	<i>Sig. (2-tailed)</i>	0.000	
	<i>N</i>	30	30

** Correlation is significant at the 0.01 level (2-tailed)

6.3.2 Case 2: Road freight costs vs GCI (World Economic Forum)

In addition to the World Bank's LPI, the Global Competitiveness Index (GCI) of the World Economic Forum was also used to determine the relationship between road freight costs and infrastructure. One of the components of the GCI is the road infrastructure index, which evaluates the road connectivity and the quality of roads among different countries.

Figure 13 shows a similar trend and clustering among comparable countries. The key difference is the much lower position of the Philippines. The Philippines ranked 134th out of 140 countries in terms of road infrastructure, even lower than most African and Latin American countries. The Philippines' rank is the second lowest among all ASEAN countries with available data (Lao PDR ranked 137th; Myanmar and Timor Leste were not included in the GCI 2018). In contrast, Thailand ranked 52nd best, while Singapore ranked first.

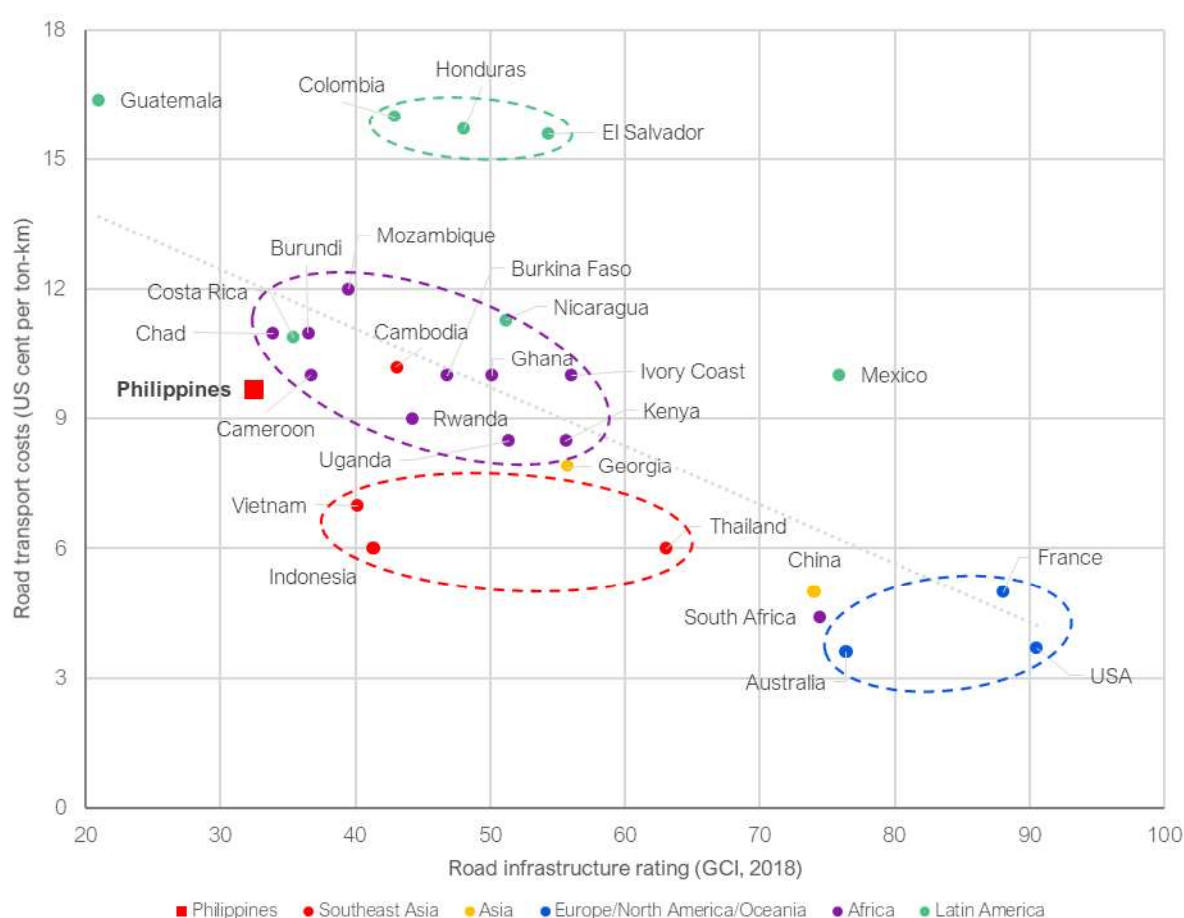


Figure 13. Relationship between road transport costs and road infrastructure rating (GCI)

Simple bivariate (Pearson's) correlation was used to determine the existence of relationship between the road freight costs and infrastructure development. The analysis, as shown in Table 7 (Case 1) and Table 8 (Case 2), yielded a moderate to high negative correlation that is statistically significant for both Case 1 ($r = -0.695$, $p < 0.001$) and Case 2 ($r = -0.656$, $p < 0.001$). This indicates that better road infrastructure leads to lower road freight costs. The moderate to high association suggests that infrastructure is only one of several factors that may have a direct effect on the transport cost. Other factors may include the following: fleet utilization, government regulation, market competition, and presence/absence of informal payments.

Table 8. Correlation between road transport costs and road infrastructure rating (GCI)

		Transport cost	Road infrastructure rating (GCI)
Transport cost	<i>Pearson Correlation</i>	1.00	-0.656**
	<i>Sig. (2-tailed)</i>		0.000
	<i>N</i>	28	28
Road infrastructure rating (GCI)	<i>Pearson Correlation</i>	-0.656**	1.00
	<i>Sig. (2-tailed)</i>	0.000	
	<i>N</i>	28	28

** Correlation is significant at the 0.01 level (2-tailed)

7. CONCLUSION

This study utilized information from a Luzon-wide survey of 36 trucking and manufacturing firms to quantify and analyze the high transport costs in the Philippines. With the country's sole reliance on roads for inland freight transport, it was deemed imperative to determine the impact of the level of road infrastructure, particularly expressways, in the trucking cost.

The results of the survey indicated that special economic zones (SEZs) were highly reliant on expressways, which are utilized in 96% of all surveyed truck routes. By comparing travel data in 2015 and 2019, the study found out that travel time generally increased among all surveyed routes, except for those where expressway development took place. In terms of the trucking cost, there was an observed increase of 1.4% per annum.

The calculated unit transport cost in 2019 was 9.67 US cent per ton-km. This was among the highest within the Southeast Asian region, along with Cambodia. Based on the study by ADB in 2012, this computed logistics cost is also outside the recommended range of logistics cost for middle-income countries which corresponds to 4 to 6 US cent/ton-km.

International benchmarking indicators such as the World Bank's LPI (Case 1) and World Economic Forum's GCI (Case 2) were used to establish the relationship between road freight costs and related infrastructure. For both cases, there is an observed clustering among countries with similar geography and/or size of economy. This paper has shown empirically that the condition of road freight transport in the Philippines is more comparable with inefficient developing country markets (e.g., landlocked African countries) than with other logistically efficient ASEAN member states (e.g., Indonesia, Thailand, and Vietnam).

Statistical analyses yielded a moderate to high negative correlation that is statistically significant for both cases. Therefore, it can be concluded that better road infrastructure lowers road freight costs.

It is worth noting for future studies that this paper did not secure disaggregate data of direct cost hence it is difficult to identify which among the direct costs is the highest and perhaps needs policy intervention. Similarly, the base year of the compared international data is not the same thus further effort may be made to secure the most updated data.

Since expressway development may be considered a long-term plan, the surveys also recommended short-term solutions. One of the most common concerns was the truck ban and the low weight limit for trucks. This called for a more synchronized and effective truck ban scheme and a revision of the anti-overloading law. Another major concern was several

obstructions on access roads, poor traffic management, and a lack of traffic discipline. It was thus recommended that stricter enforcement of traffic rules and city ordinances be done.

DTI and World Bank (2017) stated that Luzon, as the most industrialized island group in the country, benefits from lower logistics costs. The findings of this paper suggest that with all expressways having built in Luzon, the cost of trucking in Visayas and Mindanao are expected to be higher than the computed transport cost. Therefore, it would be necessary for the Philippines to invest in the development of expressways, and/or other modes of inland freight transport to enhance its international trade competitiveness.

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